CSEN604: Data Bases II

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Query Optimization II

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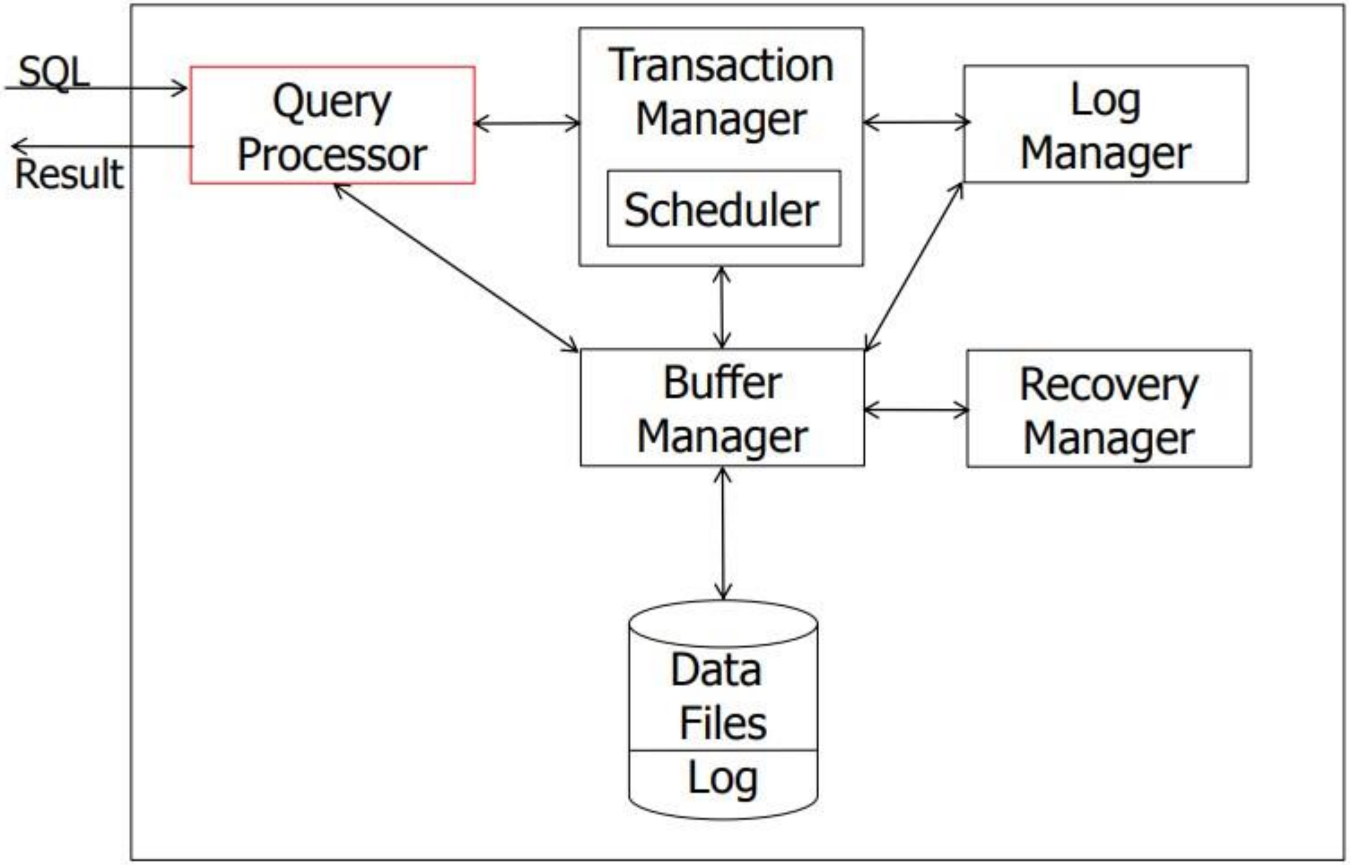
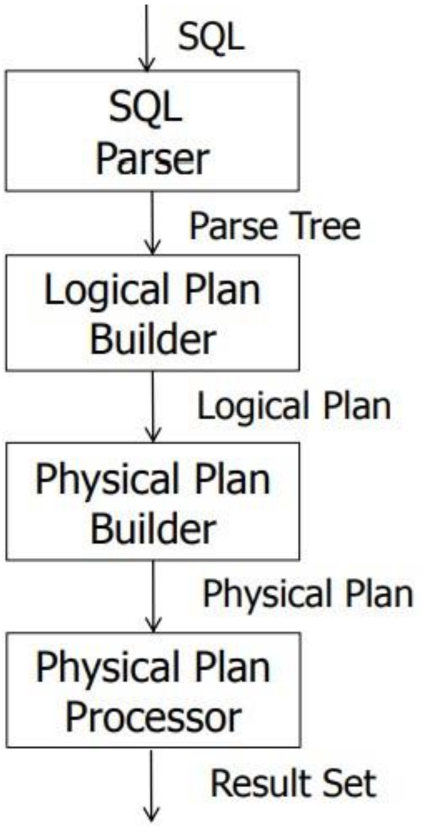
Outline

- Introduction.
- Relational Algebra/Logical Plan Optimization.
- Cost-based Optimization.

Introduction



Introduction



Introduction

StarsIn

title	starName	...	

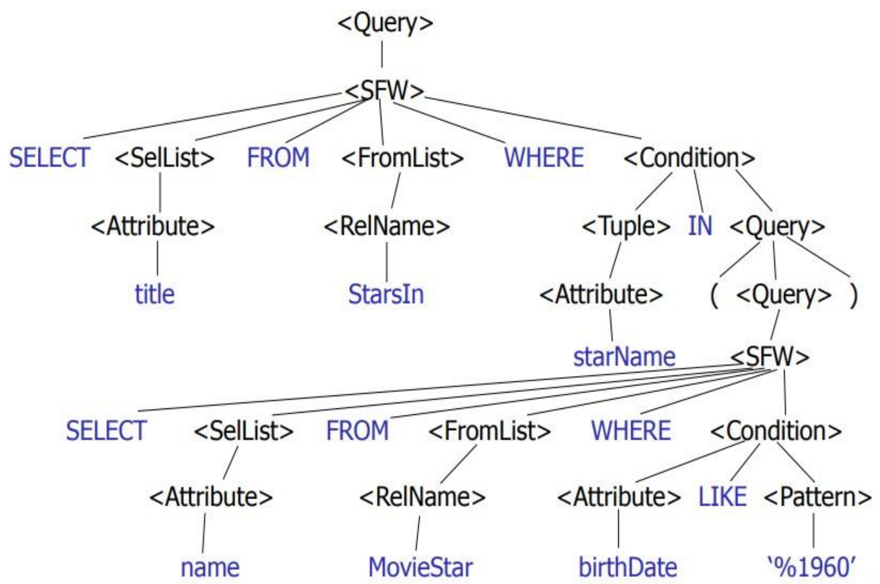
MovieStar

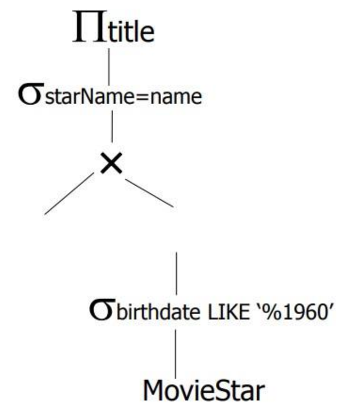
name	birthdate	...	

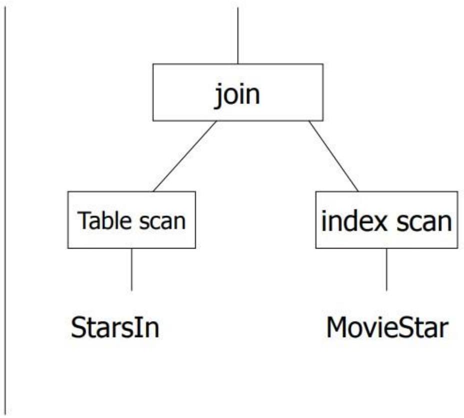
(Find the movies with stars born in 1960)

```
SELECT title
FROM StarsIn
WHERE starName IN (
     SELECT name
     FROM MovieStar
     WHERE birthdate LIKE '%1960'
);
```

Introduction







Relational Algebra/Logical Plan Optimization

- Relational Algebra is procedural language and gives you a certain order for executing a query.

- By applying certain rules, you can rearrange the order of executing the Query without changing the result.

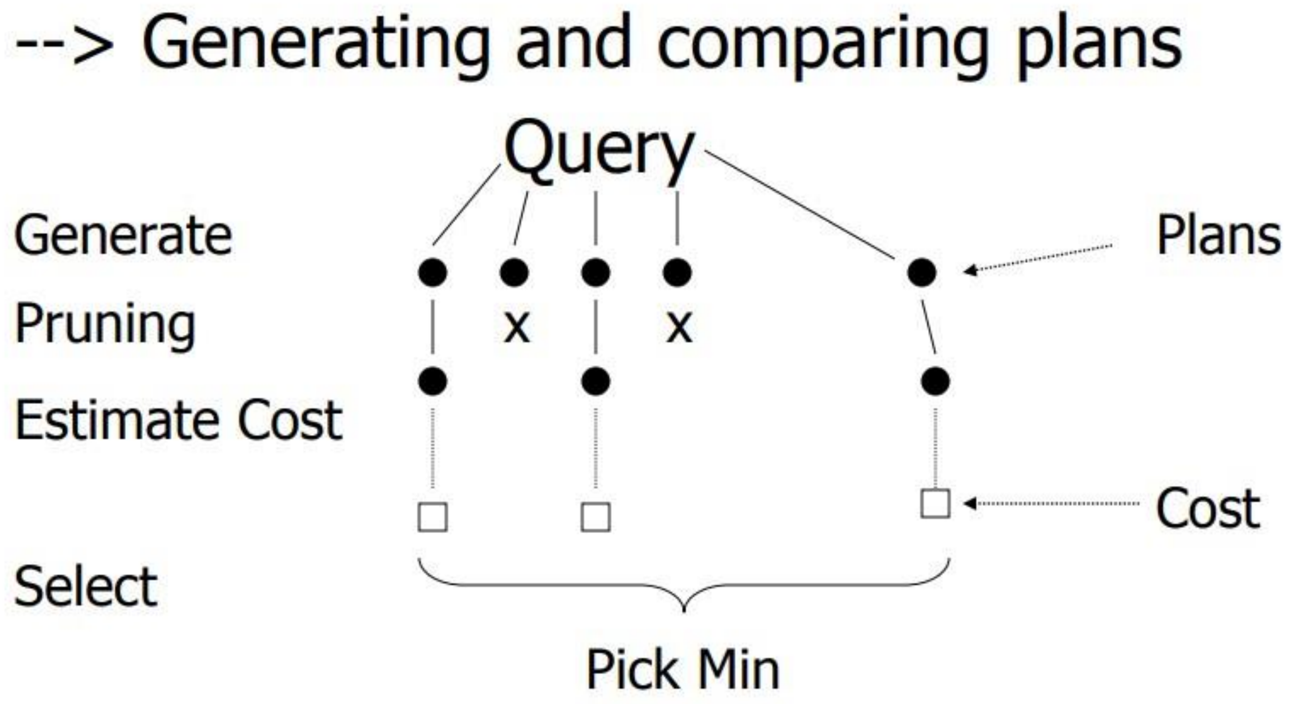
- Such reordering can incredibly improve efficiency.

- For even a simple SQL, there are several alternative relational algebra trees – all produce the same result set. One of them will be optimal.

- There are 2 types of optimizers:
  - Heuristic-based Optimizers:
    - Apply greedy rules that aims to improve plan performance.
    - Limited, not always yield optimal plan, not popular anymore, but some RDBMS still use it.
  - Cost-based Optimizers:
    - Use a cost model to estimate cost of each plan and select cheapest cost.

Cost-based Optimization

- Use a cost model to estimate cost of each plan and select cheapest cost.



Cost-based Optimization

- Statistics for Relation *R*

  - T(R) : # tuples in R
  - S(R) : # of bytes in each R tuple
  - B(R) : # of blocks (pages) to hold all R tuples
  - V(R, A): # distinct values in R for attribute A

Cost-based Optimization

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R

A	B	C	D
cat	1	10	a
cat	1	20	b
dog	1	30	a
dog	1	40	c
bat	1	50	d

A: 20 byte string
B: 4 byte integer
C: 8 byte date
D: 5 byte string

T(R) = 5
V(R,A) = 3
V(R,B) = 1

S(R) = 37
V(R,C) = 5
V(R,D) = 4

Cost-based Optimization

- Statistics for Relation R

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R

A	B	C	D
cat	1	10	a
cat	1	20	b
dog	1	30	a
dog	1	40	c
bat	1	50	d

V(R,A)=3
V(R,B)=1
V(R,C)=5
V(R,D)=4

$$W = \sigma_{z=val}(R) \quad T(W) = \frac{T(R)}{V(R,Z)}$$

$$T(\sigma_{A=val(R)}) = 5/3 = 1.6 \quad T(\sigma_{B=val(R)}) = 5/1 = 5$$

$$T(\sigma_{C=val(R)}) = 5/5 = 1 \quad T(\sigma_{D=val(R)}) = 5/4 = 1.25$$

Cost-based Optimization

- Values in select expression Z = val are uniformly distributed over possible V(R,Z) values.
- Alternate Assumption: Values in select expression Z = val are uniformly distributed over domain with DOM(R,Z) values.

Example

R

A	B	C	D
cat	1	10	a
cat	1	20	b
dog	1	30	a
dog	1	40	c
bat	1	50	d

Alternate assumption

DOM(R,A)=6

DOM(R,B)=8

DOM(R,C)=9

DOM(R,D)=15

$$W = \sigma_{Z=val}(R) \quad T(W) = \frac{T(R)}{DOM(R,Z)}$$

$$T(\sigma_{A=val(R)})=5/6=0.83 \qquad T(\sigma_{B=val(R)})=5/8=0.625$$

$$T(\sigma_{C=val(R)})=5/9=0.555 \qquad T(\sigma_{D=val(R)})=5/15=0.333$$

Cost-based Optimization

What about $W = \sigma_{z \geq val}(R)$?

$T(W) = ?$

- Solution # 1:

  $T(W) = T(R)/2$

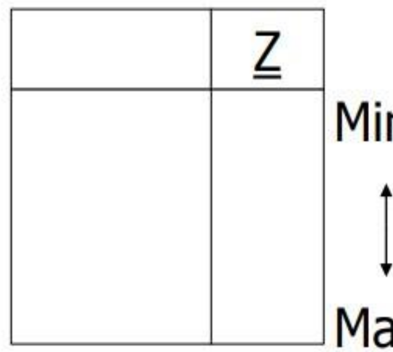
- Solution # 2:

  $T(W) = T(R)/3$

Cost-based Optimization

- Solution # 3: Estimate values in range

<u>Example</u> R



Min=1 $V(R,Z)=10$

$T(R) = 250$

Max=20 $W = \sigma_{Z \geq 15}(R)$

We are looking for a fraction f of the range

$$f = \frac{\text{matching rows}}{\text{whole range}} = \frac{\text{max-val}+1}{\text{max-min}+1} = \frac{20-15+1}{20-1+1} = \frac{6}{20} = 0.3$$

Cost-based Optimization

<u>Size estimate:</u> $W = \sigma_{z\ != \ val}(R)$

- 1/V(R,A) will fail the condition

$$T(W) = \frac{T(R)\,(V(R,a) - 1)}{V(R,a)}$$

Cost-based Optimization

Size estimate: $W = \sigma_{a=val_1 \text{ and } b<val2}(R)$

- A cascade of simple selections

- E.g.

$R(a,b,c)$

$T(R) \quad = 10{,}000$

$V(R,a) = 50$

$W = \sigma_{a=10 \text{ and } b < 20}(R)$

$$T(W) = (T(R) / V(R,A)) \times (1/3)$$
$$= (10{,}000 / 50) \times (1/3) = 67$$

Cost-based Optimization

Size estimate: $W = \sigma_{a=val_1 \text{ or } b<val_2}(R)$

- E.g. R(a,b,c)

  $T(R) = 10{,}000$

  $V(R,a) = 50$

  $W = \sigma_{a=10 \text{ or } b < 20}(R)$

  $T(\sigma_{a=10}) = T(R) / V(R,A) = 200$

  $T(\sigma_{b<20}) = T(R) / 3 = 3333$

  $T(W) = 3533$

Cost-based Optimization

- Statistics for Relations $R1 \; and \; R2$

– T(R) : # tuples in R
– S(R) : # of bytes in each R tuple
– B(R) : # of blocks (pages) to hold all R tuples
– V(R, A): # distinct values in R for attribute A

for W = R1 x R2

$$T(W) = T(R1) \times T(R2)$$

for W = R1 x R2

$$S(W) = S(R1) + S(R2)$$

Cost-based Optimization

<u>Size estimate</u> for $W = R1 \bowtie R2$

Let x = attributes of R1

y = attributes of R2

$\boxed{\text{Case 1}}$ $X \cap Y = \emptyset$

Same as R1 x R2

Cost-based Optimization

Case 2 $\quad W = R1 \bowtie R2 \qquad X \cap Y = A$

R1

A	B	C

R2

A	D

Assumption:

$V(R1,A) \leq V(R2,A) \Rightarrow$ Every A value in R1 is in R2

$V(R2,A) \leq V(R1,A) \Rightarrow$ Every A value in R2 is in R1

- $V(R1,A) \leq V(R2,A) \quad T(W) = \dfrac{T(R2)\,T(R1)}{V(R2,A)}$

- $V(R2,A) \leq V(R1,A) \quad T(W) = \dfrac{T(R1)\,T(R2)}{V(R1,A)}$

$$T(W) = \frac{T(R2)\,T(R1)}{\max\{\, V(R1,A), V(R2,A) \,\}} \qquad S(W) = S(R1) + S(R2) - S(A)$$